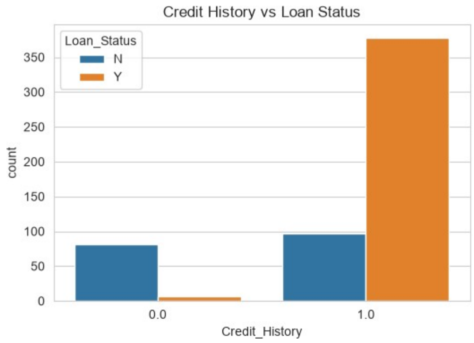
Loan Approval Prediction

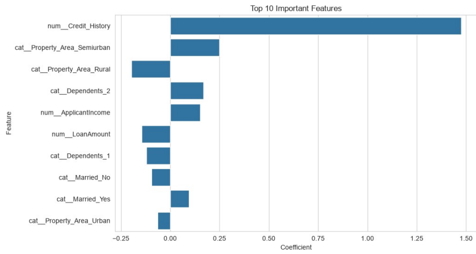
614	0.89	0.87	0.50
applicant records (69% approved / 31% rejected)	F1 score — Logistic Regression (best model)	ROC-AUC — Logistic Regression	recommended deployment threshold

Key Findings

- **Credit history dominates.** Applicants with good credit history are approved at a vastly higher rate — the largest model coefficient by far (1.47, >5x the next feature).
- **Logistic Regression wins on every metric.** F1=0.89, ROC-AUC=0.87, beating Random Forest, Gradient Boosting, and Decision Tree on this dataset.
- **Income & property area matter less.** Higher income and Semiurban property both raise approval odds; Rural property lowers it.
- **SMOTE improved recall** on the harder-to-predict Rejected class versus training on raw imbalanced data.
- **0.50 is the optimal threshold.** Raising it to 0.60/0.70 trades large recall losses for minimal precision gains — F1 falls from 0.89 to 0.78 then 0.57.



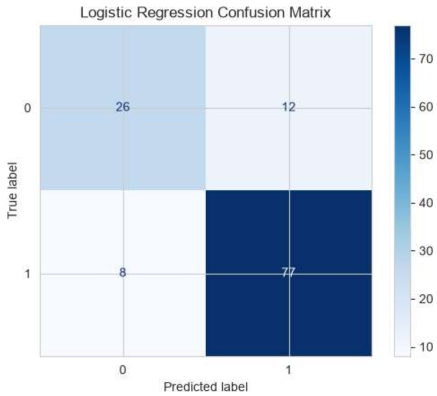
Credit history vs. approval



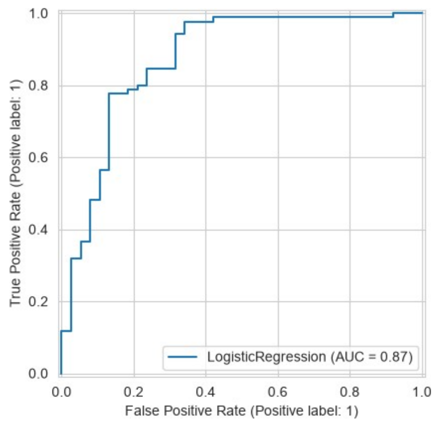
Feature importance (coefficients)

Model Comparison (Test Set)

Model	F1	ROC-AUC
Logistic Regression ■	**0.89**	**0.87**
Random Forest	0.86	0.77
Gradient Boosting	0.83	0.76
Decision Tree	0.80	0.70



Confusion matrix (test set)



ROC curve (AUC = 0.87)

Recommendation

Deploy **Logistic Regression at the default 0.50 threshold** as an initial screening tool feeding human underwriting review — top performer and most explainable, which matters for adverse-action reasoning in lending.

Source: Loan_Approval_Predicition.ipynb · loan_approval_models.pkl · best_loan_model.pkl